

# Optimizing Patient Admission Using Mathematical Programming

## Master Thesis

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# 1. Introduction

According to the so-called “Worry Barometer” of Switzerland, increasing healthcare costs remain the main worry of the population in 2025 (UBS 2026). The reason behind these concerns becomes apparent when looking at the costs of healthcare. According to the Swiss federal statistical office, the healthcare costs in Switzerland for the year 2024 amounted to 97.2 billion Swiss francs, which translates into 11.4% of the Swiss Gross Domestic Product (GDP) (Federal Statistical Office 2026*a*). Compared to the previous year, healthcare costs increased by 4.1% (Federal Statistical Office 2026*b*). These numbers highlight the big potential and need for efficiency improvement to reduce the growing expenses and underline the importance of addressing optimization strategies in this field. One big part is to plan the limited resources efficiently, to avoid a costly misallocation.

The Patient Admission Scheduling (PAS) problem introduced by Demeester et al. (2010) revolves around the assignment of patients to a hospital bed over time that best fits their individual needs while considering resource constraints and hospital policies. Manual planning of these allocations is time-consuming and tends to result in underutilization of scarce hospital resources, which motivates the use of automated decision-making tools (Turhan and Bilgen 2017). This thesis follows the decision problem presented by Turhan and Bilgen (2017). Given is a set of patients, a set of rooms, and a planning horizon. Sought is the assignment of each patient to a bed to maximize overall satisfaction. The presented problem deals with elective procedures, meaning that treatments are scheduled in advance with a fixed admission and departure date. The patients stay for at least one night, so neither emergencies nor ambulant procedures are addressed. Each patient is assigned to a room per night that best meets their needs and wishes, allowing transfers to other rooms during their stay. Due to limited capacities in the number of rooms with different properties, not all demands per patient can be satisfied at all times. Therefore, the optimal solution requires a compromise which aims to overall minimize inconveniences like room transfers, rooms with mixed genders or a room property mismatch.

Since its introduction, the PAS problem evolved from relatively simple static problem types towards larger-scale and even dynamic optimization systems. Therefore, various solution approaches coevolved from exact mathematical formulations progressing over to heuristic approaches and hybrid versions of both. Even though exact mathematical solvers can efficiently find optimal solutions for smaller-scale problems, they demand substantial computational power and are highly time-intensive for larger-scale real-world problems as faced in hospitals. Here, heuristic strategies are needed to reach good-enough solutions in a reasonable amount of time at acceptable costs. The heuristic strategies are separated

into matheuristics and metaheuristics. While matheuristics aim to break down the problem into subproblems to solve with an exact solver, metaheuristics use heuristic search strategies without exact solvers. Some examples for metaheuristics applied to the PAS problem are hyper heuristics (Bilgin et al. 2012), hybrid tabu search (Demeester et al. 2010), and simulated annealing (Ceschia and Schaerf 2011). The latter solution approach was also extended to dynamic instances accounting for uncertainty by including more real-life features like emergency patients or dynamic admission and release dates (Ceschia and Schaerf 2012). Popular solver-based approaches include column generation (Range et al. 2014), fix-and-optimize heuristics (Turhan and Bilgen 2017), and further advanced approaches to the exact MIP solver (Bastos et al. 2019) (Liu et al. 2024).

Since matheuristics provide good solutions for large scale problems by combining the scalability of metaheuristics with the optimal problem solving of mathematical programming, we propose two different fix-and-optimize matheuristics following the idea of Turhan and Bilgen (2017). We aim to find feasible and reasonably good solutions within a time limit of one hour for the twelve benchmark datasets available on the PAS website (Demeester, Peter 2009). In the first matheuristic approach, the Adaptive Cohort Fix-and-Optimize Matheuristic, the subproblem to be optimized per iteration is defined by choosing a unique set of worst admission dates based on patients' penalties. A cohort is formed by grouping all patients arriving on these specific days. The patients in this cohort are then left open for optimization as all other patients are fixed to their current assignments. After a set of iterations, patient penalties are recalculated, and the time window of the subproblem is expanded. Compared to the complete MIP model, the Adaptive Cohort Fix-and-Optimize Matheuristic produces better solutions in the majority of instances. Especially for the bigger instances, where the exact model would need more time or computational power to even start the optimization, this matheuristic clearly yields better results. The second approach, a Multi-Strategy Fix-and-Optimize Matheuristic, defines the subproblem per iteration through one of three complementary decomposition strategies. Room clustering groups patients of the same department, time-window decomposition selects patients whose stay intersects a sliding window over the planning horizon, and swap chain builds chains of mutually blocking patients via breadth-first search. The three strategies alternate: when one fails to improve the incumbent, the next is activated, and the procedure terminates after three full passes without further improvement. Compared to the complete MIP model, the Multi-Strategy Fix-and-Optimize Matheuristic reaches a solution within twenty percent of the final base model objective on all twelve benchmark instances, taking under four minutes on each and roughly one minute on average. On several large instances, where the base model cannot close its optimality gap within the time limit, the matheuristic returns a lower final objective.

In Chapter 2, we explain the decision situation in more detail. In Chapter 3, we summarize the current state of the literature addressing solver-free and solver-based approaches for the PAS problem. In Chapter 4, we present the notation and formulation of the base model from Turhan and Bilgen (2017). In Chapter 5 and 6, we describe the developed matheuristics and their application on an illustrative example. In Chapter 7, we present the experimental setup alongside the results. In Chapter 8, the thesis is concluded.

## 2. The Patient Admission Scheduling Problem

In this chapter, we describe the decision-making situation considered in this thesis: the allocation of elective patients to hospital rooms over a specific planning horizon. This planning problem is known in the literature as the PAS problem. In Section 2.1, we describe the problem from the perspective of hospital management. In Section 2.2, we present a small example that illustrates the planning problem.

### 2.1 Problem Statement

The decision-making situation considered here arises during the daily admission planning of a hospital. Elective patients, i.e. patients whose admission is known in advance, and rooms must be allocated for the entire duration of their stay. When making these allocations, hospital planners must take into account the medical needs of each patient, the spatial conditions of the available rooms, and a range of preferences which, whilst not strictly necessary, contribute significantly to patient satisfaction and the smooth running of the wards.

A fixed planning horizon is specified for the planning problem. The horizon consists of a sequence of consecutive nights, with one night representing the smallest unit of time in the schedule. The hospital is divided into a number of departments. Each department specializes in the treatment of specific clinical conditions, such as cardiology, pediatrics, or general surgery, and offers one or more specialist areas. Each department comprises a number of rooms. Each room has a capacity that determines how many patients can share the room on the same night. Furthermore, each room has individual characteristics: it may be reserved for a specific gender or open to both genders, it may restrict admission to patients within a specific age group, and it is equipped with specific medical facilities, such as an oxygen supply or telemetry monitoring.

A group of patients is also specified. For each patient, the night of admission and the length of stay are known in advance. Each patient requires treatment in a specific medical speciality, has a known gender and age, and has individual preferences regarding room type (single, twin, or shared room). Some patients require, for medical reasons, a room that is equipped with specific medical facilities, whilst other features are merely preferred but not strictly necessary.

The objective is to assign each patient to exactly one room on every night of the stay such that all hard constraints are satisfied and the weighted penalties for violating patient-

related soft constraints are minimized. The following constraints must be satisfied for every valid allocation:

- Each patient is allocated exactly one room for every night of their stay.
- The admission and discharge nights for each patient are fixed and cannot be changed.
- On any given night, the number of patients sharing a room does not exceed the capacity of that room.

Furthermore, the allocation should take into account a set of soft constraints. Each preference is associated with a penalty that quantifies the cost of a violation. The following preferences are taken into account:

- *Specialism of the department*: a patient should be treated in a department whose specialism corresponds to the medical need.
- *Specialism of the room*: within a department, certain rooms are dedicated to particular specialisms. Patients should be placed in a room matching their specialism.
- *Gender policy*: a patient should be placed in a room whose gender policy is compatible. If a no-mixing room hosts patients of different genders on the same night, a gender mixing penalty is incurred.
- *Age policy*: a patient should be placed in a room whose age policy is compatible with the age of the patient.
- *Mandatory room features*: the room should be equipped with the medical features that the treatment requires.
- *Preferred room features*: the room should also be equipped with features that are not strictly necessary but are preferred for the comfort or recovery of the patient.
- *Room type*: the type of the assigned (single, twin, or shared room) should match the preference of the patient.
- *Transfers*: a patient should stay in the same room for the entire duration of the stay. Each change of room during a stay is undesirable.

The objective is to find an assignment of patients to rooms that satisfies all of the soft constraints listed above, and that minimizes the total penalty incurred by violations of the soft constraints. Since the preferences differ in importance, each soft constraint is weighted individually. A violated gender policy, for instance, weighs more heavily than a missing preferred feature. The weights used in this thesis follow the convention adopted by Turhan and Bilgen (2017).

## 2.2 Illustrative Example

In order to illustrate the planning problem described in Section 2.1, we introduce a small instance of the PAS problem. The instance is small enough to be discussed in detail, but it contains all the elements of the problem. The example is inspired by the toy instance of Turhan and Bilgen (2017) and adapted to highlight the key tradeoffs between medical needs, capacity, and preferences.

The instance comprises a planning horizon of five consecutive nights, two departments, three rooms, and five elective patients. Table 2.1 lists the two departments and their major specialisms.

| Department | Major specialism | Identifier |
|------------|------------------|------------|
| Cardiology | Cardiology       | $d_1$      |
| Pediatrics | Pediatrics       | $d_2$      |

Table 2.1: Departments of the illustrative instance and their major specialisms.

The characteristics of the three rooms are summarized in Table 2.2. Rooms  $r_1$  and  $r_2$  belong to the cardiology department. Room  $r_2$  is a single room equipped with both oxygen supply and telemetry, while room  $r_1$  is a twin room with oxygen supply only. Room  $r_3$  belongs to the pediatrics department, is a twin room, and admits only patients up to the age of sixteen.

| Room  | Department | Capacity | Gender policy | Age limit | Features          |
|-------|------------|----------|---------------|-----------|-------------------|
| $r_1$ | $d_1$      | 2        | mixed         | none      | oxygen            |
| $r_2$ | $d_1$      | 1        | mixed         | none      | oxygen, telemetry |
| $r_3$ | $d_2$      | 2        | mixed         | $\leq 16$ | oxygen            |

Table 2.2: Rooms of the illustrative instance with their characteristics.

The five patients in the instance are described in Table 2.3. For each patient, the table reports the gender, the age, the required specialism, the preferred room type, the medically required equipment, the admission night (Adm. night), and the length of stay in nights (LoS). None of the patients in this small instance has a wish for additional preferred equipment.

With this data, the planner faces the following situation. Three beds are available in the cardiology department (room  $r_1$  with capacity two and room  $r_2$  with capacity one), and two beds are available in the pediatrics department (room  $r_3$  with capacity two). On the busiest nights, four of the five patients are present simultaneously, so the capacity is almost fully utilized and the room choices are tightly coupled. Patient  $p_2$  is the only patient who requires telemetry and at the same time prefers a single room, which makes room  $r_2$  the natural choice for him. Since room  $r_2$  has capacity one, patient  $p_2$  blocks it for the entire duration of his stay. Patients  $p_4$  and  $p_5$  both require treatment in the pediatrics department and are both within the age limit of room  $r_3$ , so they can share the

| Patient | Gender | Specialism | Room pref. | Required eq. | Age | Adm. night | LoS |
|---------|--------|------------|------------|--------------|-----|------------|-----|
| $p_1$   | F      | Cardiology | twin       | oxygen       | 55  | 1          | 2   |
| $p_2$   | M      | Cardiology | single     | telemetry    | 62  | 1          | 3   |
| $p_3$   | M      | Cardiology | twin       | oxygen       | 70  | 3          | 2   |
| $p_4$   | F      | Pediatrics | twin       | oxygen       | 8   | 2          | 3   |
| $p_5$   | F      | Pediatrics | twin       | oxygen       | 12  | 1          | 3   |

Table 2.3: Patients of the illustrative instance with their characteristics. *Adm. night* denotes the admission night, and *LoS* the length of stay in nights.

only available pediatrics room. Patients  $p_1$  and  $p_3$  both prefer a twin room in cardiology, but their stays do not overlap, which allows the planner to assign both of them to room  $r_1$  without violating the capacity. Finding an assignment that respects all of the rules listed in Section 2.1 and, at the same time, keeps the total penalty as low as possible is the central planning task of the PAS problem. This example is small, but the problem becomes considerably more complex as the size of the instance increases.



## 3. Literature Review

The literature on the Patient Admission Scheduling problem spans two broad strands: solver-free metaheuristics that operate directly on the solution space, and solver-based approaches that exploit an explicit integer-programming formulation. In Section 3.1, we present the former, covering tabu search, hyper-heuristics, simulated annealing, and related methods. In Section 3.2 we turn to the latter, reviewing the column-generation, direct MIP, and MIP-heuristic contributions that represent the current state of the art. Key contributions are discussed with respect to the decision situation considered, the solution approach proposed, and the main results reported. The section concludes by identifying the research gap that motivates this thesis.

### 3.1 Solver-free Solution Approaches

While exact mathematical approaches like mixed integer programming (MIP) can be applied to solve the PAS problem to a guaranteed optimal solution, metaheuristics were introduced early on to spare resources. These approaches are designed to explore large solution spaces in the cases where exact optimization is too slow and aim for good solutions in a reasonable amount of time (Talbi 2009). Upon introduction of the static PAS problem by Demeester et al. (2010), the first metaheuristic solution approach presented in this paper is a hybrid tabu search. The tabu search solution approach is a hybrid version of local search with a tabu search algorithm. Local search is the basis candidate method for optimization, which requires three essential components: a solution, the neighborhoods describing reachable solutions within a single move, and a cost function for the quality measure of the solution (Demeester et al. 2010). The strategy of tabu search, already described by Glover (1986), follows the concept of local search but adds memory to the search process. The idea is to forbid moves recently taken, i.e. add them to a tabu list in the memory, to prevent reversing moves and cycling within local minima. The strategy operates on the assumption that choosing a poor move has no value except insofar as it prevents revisiting an already explored path (Glover 1986). Specifically, during the iterative search procedure, neighborhoods of the current solution are explored through patient reassignments and room swaps while maintaining a tabu list. The tabu list shortens when a better solution is found, and vice versa, extends when no better solution is found to force the exit of a local minimum. The presented tabu search algorithm is furthermore improved by a token-ring approach, where neighborhoods are switched after a number of

non-improving moves (Demeester et al. 2010).

Ceschia and Schaerf (2011) describe another approach, applying simulated annealing. The simulated annealing approach originates from physical annealing processes when heated materials cool down and arrange to reach a low-energy stable state (Kirkpatrick et al. 1983). Applied to an optimization strategy, they implement a cooling rate, deflecting the decrease in acceptance probability of worse neighbor solutions after a move. They create neighborhood solutions by randomly applying moves to change rooms and swap patients, and they are graded by a cost function. In principle, in the early phase (high temperature) worse solutions are accepted more often boosting exploration while later on (lower temperature) current solutions are exploited by accepting mostly moves that lead to improvements. This approach results in better outcomes in terms of quality and computational time than previously described work using hyper-heuristics, thanks to the new neighborhood combinations. Furthermore, Ceschia and Schaerf (2011) provide an application of the simulated annealing approach to dynamic cases, where patient admission and discharge dates only become available during the planning period itself, demonstrating a two day forecast is enough to construct a good quality scheduling. Follow-up work by Ceschia and Schaerf (2012) further explores the expansion of the PAS to account for uncertainty (PASU to include Uncertainty), including emergency patients, uncertainties regarding the length of stay, and delayed patient admission. This expansion increases the practical applicability of the solution approach by bringing the model closer to reality. They apply a similar approach previously described, but with adaptations like allowing additional shifts for the neighborhood, shifting a patient's admission by one day backward or forward. A similar approach is the Great Deluge algorithm originally developed by Dueck (1993), but rather than accepting worse solutions to a certain probability, the algorithm accepts all solutions below a certain cost level (so-called water level). The application of the non-linear great deluge algorithm to the static PAS problem is proposed by Kifah and Abdullah (2015). Here, the water level decrease is non-linear and depends on the search performance in a way that the water level can also increase temporarily to escape local minima. Similarly, the late acceptance hill climbing (Burke et al. 2008) applied to the static PAS problem by Bolaji et al. (2018), escapes local minima by comparing new candidate solutions to stored solutions generated several iterations before.

A broader view on solving the PAS problem is the application of a hyper-heuristic, presented by Bilgin et al. (2012). In contrast to metaheuristics, hyper-heuristics deploy a set of low-level heuristics instead of operating directly on the solution set. The term hyper-heuristic broadly defines the approach of selecting one low-level heuristic from a set in every iteration and applying it to the candidate solution. The resulting solution is then either accepted or rejected according to a move acceptance criterion. Both the heuristic selection method and the move acceptance criterion can be based on (meta-) heuristics. In this paper, they assess combinations of selection methods (simple random, choice function, or dynamic heuristic set with simple random (DHS)) and move acceptance criteria (only improving, improving and equal, simulated annealing (Kirkpatrick et al. 1983), and great deluge (GD) (Dueck 1993)). As a result, they observe that the DHS-GD-hyper-heuristic outperforms the others and that the performance of the hyper-heuristic is dominated by

the choice of the move acceptance criteria (Bilgin et al. 2012).

Instead of choosing the neighborhoods randomly to create new solutions, Lusby et al. (2016) use adaptive large neighborhood search in a simulated annealing approach. The paper treats a dynamic patient admission scheduling problem where the patients arrive constantly, therefore, the planning has to be changed continuously. The neighborhood is created with a destroy and repair mechanism, instead of random moves. The destruction and repair mechanism follows different methods that are initially equally likely to be chosen. The weights of the different methods are updated after observing their performance in 100 iterations, adapting towards better-performing methods in later phases.

## 3.2 Solver-based Solution Approaches

The approaches discussed here, delegate core optimization work to a MIP solver, either by passing the full model directly or by embedding the solver in a decomposition-based heuristic.

An early dedicated optimization-based solver approach to PAS is the column-generation heuristic of Range et al. (2014). The decision situation is a deterministic PAS variant in the spirit of Demeester et al. (2010): elective patients with known admission and discharge dates must be assigned to rooms over a fixed planning horizon, subject to medical, gender, capacity, and patient-preference requirements. The solution approach reformulates the problem as a set-partitioning master problem in which each variable represents a complete room schedule. The associated room-scheduling subproblems serve as the pricing problems in a column-generation-based optimization heuristic that also uses dynamic constraint aggregation and branch-and-bound ideas. The paper is among the early works to address PAS with an optimization-based approach built on a strong reformulation rather than a pure metaheuristic. On the benchmark instances of Demeester et al. (2010), the approach yields new best-known solutions for six of the twelve instances and produces the tightest lower bounds reported up to that date. The substantially higher computation times compared to later heuristic methods, however, restrict the approach effectively to the smaller benchmark instances.

The paper most directly relevant to this thesis is Turhan and Bilgen (2017). The decision situation is a deterministic PAS formulation based on the preprocessing and reformulation of Ceschia and Schaerf (2011). Although it is closely related to the variant considered by Range et al. (2014), the treated constraint variants differ. The solution approach consists of two MIP-based heuristics that are applied sequentially: a fix-and-relax (F&R) procedure constructs an initial feasible solution by solving a sequence of reduced MIPs under time and patient decompositions, where the time windows are not limited to a single night, and a subsequent fix-and-optimize (F&O) procedure iteratively improves the incumbent by re-optimizing small subsets of binary variables defined either by time windows or by patient groups. What is new in the paper is twofold: it introduces a MIP-based heuristic technique to PAS, and it shifts the trade-off between solution quality and computation time established by the earlier metaheuristic and exact approaches. The reported results

illustrate this shift: on the benchmark instances of Demeester et al. (2010), the method achieves objective values within five to fifteen percent of the best-known solutions for most instances in under three minutes of computation time. This is often substantially faster than the earlier approaches of Range et al. (2014) and Ceschia and Schaerf (2011), but the magnitude of the speed-up is instance-dependent and therefore cannot be described as a uniform improvement by one or two orders of magnitude.

Bastos et al. (2019) return to the deterministic PAS variant and pose the question of the extent to which advances in MIP-solver technology since the introduction of the problem allow it to be solved directly. The solution approach is a comprehensive, exact MIP formulation that retains all restrictions of the original model of Demeester et al. (2010), in contrast to simplified or preprocessed approaches, and is solved with a warm-start strategy based on a simplified no-transfer model. What is new in the paper is the demonstration that a direct MIP solve, made tractable by a comprehensive exact formulation and modern solver technology, can provide a competitive alternative to earlier dedicated optimization approaches. The approach yields nine new best-known solutions out of thirteen instances and proves optimality for several of the best-known solutions. For the two largest instances, however, no improvement was obtained within a time budget of 24 hours.

The most recent solver-based contribution is Liu et al. (2024). The decision situation is again the deterministic PAS of Demeester et al. (2010). The solution approach extends the model of Ceschia and Schaerf (2011) by a constraint-aggregation scheme: The gender policy and patient transfer constraints are replaced by compact aggregate counterparts, reducing the size of the integer program that the solver must handle. The resulting reduced model is then solved in a two-stage optimization procedure. What is new in the paper is the systematic exploitation of constraint structure to shrink the formulation, which substantially improves the solver’s ability to find optimal solutions. For the standard PAS problem, the approach yields six new best-known solutions, one of them with proven optimality, and reaches the already-known optima faster on five further instances, improving on several of the results reported by Bastos et al. (2019).

Taken together, the solver-oriented PAS literature combines optimization-based heuristics, such as column generation and MIP-based matheuristics, with exact MIP approaches based on stronger formulations and model reduction. These have been driven forward primarily by improvements in formulation strength and in commercial solver technology, and now establish best-known solutions on most small and medium instances. For the largest benchmark instances, however, proving optimality remains challenging within practically relevant time budgets.

Within the literature reviewed here, no study was identified that directly benchmarks a contemporary reimplement of the F&O idea of Turhan and Bilgen (2017) against a direct MIP solve of the same formulation. This question forms the starting point of Chapters 5 and 6, in which we use such a matheuristic precisely in order to take advantage of the properties of both exact and heuristic approaches.

## 4. A Mixed-Integer Linear Programming Model

This chapter presents the optimization model that serves as the formal foundation of this thesis. The model describes the PAS problem introduced in Chapter 2 as a mixed-integer linear program. We adopt the formulation of Turhan and Bilgen (2017), which is itself based on the reformulation of the PAS problem proposed by Ceschia and Schaerf (2011). Section 4.1 presents the notation, and Section 4.2 states the model formulation and explains its objective function and constraints.

### 4.1 Notation

Table 4.1 lists all sets, parameters, and decision variables used in the model. Each patient  $p$  has a fixed admission date  $AD_p$  and a fixed discharge date  $DD_p$ . The patient occupies a bed on every night of the resulting interval, which we collect in the day set  $D_p$ . The parameter  $C_{p,r}$  aggregates all room-assignment penalties that depend only on the pairing of a patient with a room. It covers the preferences (soft constraints) introduced in Chapter 2, each weighted according to its relative importance, and is computed once per patient–room pair before solving the model. The gender-policy penalty and the transfer penalty are kept separate from  $C_{p,r}$  because they depend on the joint assignment of multiple patients and on consecutive-day decisions, respectively, and therefore cannot be pre-computed per patient–room pair.

|                   |                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sets</b>       |                                                                                                                                          |
| $p \in P$         | Set of patients.                                                                                                                         |
| $p \in P_M$       | Set of male patients, $P_M \subseteq P$ .                                                                                                |
| $p \in P_F$       | Set of female patients, $P_F \subseteq P$ .                                                                                              |
| $d \in D$         | Set of days.                                                                                                                             |
| $d \in D_p$       | Set of days on which patient $p$ occupies a bed, $D_p = \{AD_p, \dots, DD_p-1\} \subseteq D$ .                                           |
| $r \in R$         | Set of rooms.                                                                                                                            |
| <b>Parameters</b> |                                                                                                                                          |
| $c_r$             | Capacity of the room $r$ .                                                                                                               |
| $AD_p$            | Admission date of the patient $p$ .                                                                                                      |
| $DD_p$            | Discharge date of the patient $p$ .                                                                                                      |
| $C_{p,r}$         | The penalty of assigning patient $p$ to room $r$ . All the room penalties are blended into this value except gender policy and transfer. |
| $w_{RG}$          | Weight of room gender policy constraint.                                                                                                 |
| $w_{Tr}$          | Weight of transfer constraint.                                                                                                           |
| <b>Variables</b>  |                                                                                                                                          |
| $x_{p,r,d}$       | = 1 if patient $p$ is assigned to room $r$ in day $d$ , 0 otherwise.                                                                     |
| $t_{p,r,d}$       | = 1 if patient $p$ is transferred from room $r$ in day $d$ , 0 otherwise.                                                                |
| $f_{r,d}$         | = 1 if there is at least one female patient in room $r$ in day $d$ , 0 otherwise.                                                        |
| $m_{r,d}$         | = 1 if there is at least one male patient in room $r$ in day $d$ , 0 otherwise.                                                          |
| $b_{r,d}$         | = 1 if there are both male and female patients in room $r$ in day $d$ , 0 otherwise.                                                     |

Table 4.1: Sets, parameters, and decision variables of the model.

## 4.2 Model Formulation

Using the notation of Section 4.1, the PAS problem is formulated as the following mixed-integer linear program. The objective function (1) minimizes the total penalty associated with assigning patients to rooms for the duration of their stay.

$$\text{Minimize} \quad \sum_{p \in P, r \in R, d \in D} C_{p,r} \cdot x_{p,r,d} + \sum_{r \in R, d \in D} w_{RG} \cdot b_{r,d} + \sum_{p \in P, r \in R, d \in D} w_{Tr} \cdot t_{p,r,d} \quad (1)$$

It is subject to the following constraints:

$$\sum_{r \in R} x_{p,r,d} = 1 \quad \forall p \in P, d \in D_p \quad (2)$$

$$\sum_{p \in P} x_{p,r,d} \leq c_r \quad \forall r \in R, d \in D \quad (3)$$

$$f_{r,d} \geq x_{p,r,d} \quad \forall p \in P_F, r \in R, d \in D \quad (4)$$

$$m_{r,d} \geq x_{p,r,d} \quad \forall p \in P_M, r \in R, d \in D \quad (5)$$

$$b_{r,d} \geq m_{r,d} + f_{r,d} - 1 \quad \forall r \in R, d \in D \quad (6)$$

$$t_{p,r,d} \geq x_{p,r,d} - x_{p,r,d+1} \quad \forall p \in P, r \in R, d \in D_p, d < DD_p - 1 \quad (7)$$

$$x_{p,r,d} \in \{0, 1\} \quad \forall p \in P, r \in R, d \in D \quad (8)$$

$$t_{p,r,d} \in \{0, 1\} \quad \forall p \in P, r \in R, d \in D \quad (9)$$

$$f_{r,d} \in \{0, 1\} \quad \forall r \in R, d \in D \quad (10)$$

$$m_{r,d} \in \{0, 1\} \quad \forall r \in R, d \in D \quad (11)$$

$$b_{r,d} \in \{0, 1\} \quad \forall r \in R, d \in D \quad (12)$$

The objective function (1) consists of three terms. The first term refers to the cost of assigning patients to rooms through the combined penalty  $C_{p,r}$  of the preference. The second term addresses the cost of violating the room gender policy, and the third term describes the cost associated with a transfer.

Constraint (2) enforces every patient to be assigned to a room on each day between the admission and the discharge date. Constraint (3) ensures that the number of patients assigned to a room on a given day cannot exceed the capacity of that room. The auxiliary variables  $b_{r,d}$ ,  $m_{r,d}$ ,  $f_{r,d}$  and  $t_{p,r,d}$  all depend on the assignment variables  $x_{p,r,d}$ , which describe the actual search space. If there is a female patient in a room, constraint (4) forces the auxiliary variable  $f_{r,d}$  to be equal to one to reflect the female presence in that room. A similar approach is taken in constraint (5) to reflect the presence of a male patient. If patients of both genders occupy the same room, constraint (6) ensures that  $b_{r,d}$  becomes one, so that the gender penalty is reflected accordingly in the objective value. Finally, constraint (7) detects whether a patient has a room change between two consecutive days. In that case, the auxiliary variable  $t_{p,r,d}$  becomes one, and the transfer

penalty is added to the objective value. Constraints (8)–(12) restrict all decision variables to binary values.

The model can be solved directly with a general-purpose MILP solver. However, the number of binary assignment variables grows with the product of the number of patients, the number of rooms, and the length of the planning horizon. For the larger benchmark instances, this renders the proof of optimality within an acceptable time budget intractable, which motivates the use of the matheuristics developed in Chapters 5 and 6.



## 5. Adaptive Cohort Fix-and-Optimize Matheuristic (Szabo)

An exact optimization of the complete model described in Chapter 4 needs a lot of time and computational power to solve, especially for large instances. We propose a matheuristic approach based on a simple decomposition strategy to address this issue. In Section 5.1, we present a general overview and the ideas behind the approach. In Section 5.2, we describe the underlying subproblems in more detail. In Section 5.3, we explain the matheuristic with an illustrative example.

### 5.1 Overview of Matheuristic

Inspired by the approach of Turhan and Bilgen (2017), we develop a matheuristic based on a fix-and-optimize strategy. This approach breaks down the static PAS problem by a decomposition strategy to allow iterative improvements by consecutively optimizing the defined subproblems using a mathematical solver. In every iteration, while the subproblem stays relaxed to enable an optimization, the remaining part of the problem is fixed to its current assignment. The strategy includes the construction of a feasible initial solution, followed by a decomposition strategy based on patient penalties, and expansions that enlarge the subproblem size later on.

We construct the initial solution as simple as possible while preventing two main sources of high penalties, namely, avoiding mixed-gender rooms and transfers. In particular, patients are greedily assigned to a room matching their gender for their whole length of stay. This enables us to find a feasible solution fast and to fill rooms up to their capacity, while leaving rooms completely empty for early improvements, although at the cost of a higher initial objective function value.

The matheuristic continuously improves the initial solution by optimizing the relaxed subproblems iteratively. Since the improvements of the assignment are directly based on the decrease of the total penalty value, we define the individual patient penalties as the base of our decomposition strategy to define the subproblems. This approach doesn't require additional time-consuming calculations, as the patient penalties are readily available. In contrast to aggregated penalties, an individual penalty is more likely to improve to the extent that a patient drops out of the worst patient subset. This leads to increased variation in the investigated subproblems and helps to avoid being stuck

in a local minimum. Since we base the decomposition strategy on the penalty value of individual patients, we build an admission date cohort of patients arriving on the same day, as they compete for the same resources and therefore need to be optimized together. To identify the worst-performing admission date subsets, patients are ranked in descending order of their individual penalties. The matheuristic then identifies a subset of unique admission dates to evaluate. After each iteration, the next lower-ranked admission dates are considered for the subsets until a defined part of the planning horizon has been explored. Then, the algorithm recalculates patient penalties and ranks them based on the current solution to identify the new worst admission date subsets for the next set of iterations.

This strategy alone can lead to local minima, where the worst patients stay the same, causing cycling around the same subproblem because the fixed patients around the cohort block any improvements. Therefore, we implement admission window expansions to relax patients with adjacent admission dates. This prevents optimizing the exact same subproblem multiple times. Even if the highest penalty patients stay the same, the problem expansion leads to new combinations to optimize. However, the expansions also lead to bigger subproblems that need higher computational time to optimize. To ensure that the matheuristic continues to run through a high number of iterations instead of optimizing one single complex subproblem exactly, we limit the time per iteration.

Algorithm 1 summarizes the steps of the procedure and shows the iterative scheme in a simplified manner. The individual parameters are explained in Section 5.2.

---

**Algorithm 1** Adaptive Cohort Fix-and-Optimize Matheuristic

---

- 1: Start global timer
  - 2: Create initial solution
  - 3: best solution  $\leftarrow$  initial solution; best obj  $\leftarrow$  initial obj
  - 4: Compute cohort size (5% of planning horizon)
  - 5: expansion level  $\leftarrow$  0
  - 6: **while** global timer < global time limit **do**
  - 7:   **for** expansion level **do**
  - 8:     Compute time limit per iteration
  - 9:     Compute penalties per patient; rank admission dates based on penalties
  - 10:    define admission date cohorts; Current cohort  $\leftarrow$  highest rank admission dates
  - 11:    **while** investigated admission days in cohorts < 50% planning horizon **do**
  - 12:      solve iteration with current cohort
  - 13:      best solution  $\leftarrow$  current solution; best obj  $\leftarrow$  current obj
  - 14:    Investigate next cohort
  - 15:    Increase expansion level
  - 16: **return** best solution; best obj
-

## 5.2 Subproblems

To start, we construct the initial solution using a greedy insertion, which means assigning a patient to the first room available. Patients are sorted by their length of stay (LoS) in a decreasing manner and added one by one to the room with the lowest room number, which is available for their whole LoS. This way, we prevent transfer penalties in the initial solution, which are most costly, and the longest stays are placed first to plan the shorter stays around them. Furthermore, the patients are only assigned to a room if their assignment does not produce a gender-mixed room, hence avoiding penalties for violating gender policies. The goal of this strategy is to find a feasible initial solution in a short amount of time. The resulting assignment includes both tightly packed and empty rooms, offering improvement potential in the early iterations. After finding the initial solution, the matheuristic decomposes the problem based on the individual patient penalties. Therefore, the matheuristic calculates the penalties per patients with the current solution before every set of iterations. The patients are then sorted according to their individual penalty in descending order, and their admission dates are selected to address critical time windows. For each iteration, the matheuristic defines a subset of unique admission dates depending on the cohort size  $CS$ , which is a percentage of the planning horizon, to account for differently sized instances. If multiple patients with high penalties shared the same admission dates, the matheuristic proceeds down the sorted patient list and adds additional admission dates to avoid duplicates, ensuring that the set size remains constant throughout the process. From these admission date subsets, we derive a cohort of patients sharing these admission dates, which are then relaxed for exact mathematical optimization. For each set of iterations, the admission date subsets are sequentially explored until the planning limit  $PL$ , which is a percentage of the planning horizon, has been investigated. This way, we ensure that several optimizations are performed before patient penalties are recalculated based on the current solution, which then form the basis for the next set of iterations. Additionally at this process step, the admission windows are expanded to avoid reinvestigating the same subproblems after recalculating patient penalties and allowing the matheuristic to escape local minima. This adds diversification to the subproblems.

For this experiment, we specify the expansion windows illustrated in Table 5.1. In the first expansion, the following day of each admission date is added, including all patients with the same admission date  $AD_p$  of highest penalty patients as well as patients with admission date  $AD_p + 1$  into the optimization. The second expansion additionally includes all patients with admission dates one day before ( $AD_p - 1$ ). The following expansions add both, one more day before and one more day after the current admission window, respectively. With a bigger global time limit or more computational power, this parameter could be adapted.

Since the idea of a matheuristic is to quickly solve many smaller subproblems instead of taking a long time solving a single more complex subproblem, we set a time limit per iteration. Due to the dynamic growth of subproblems with the expansion levels, the solver needs progressively more time to effectively start an optimization. Hence, the time limit

| Expansion | Admission dates             |
|-----------|-----------------------------|
| 0         | $AD_p$                      |
| 1         | $AD_p, AD_p + 1$            |
| 2         | $AD_p - 1, AD_p, AD_p + 1$  |
| 3         | $AD_p - 2, \dots, AD_p + 2$ |
| 4         | $AD_p - 3, \dots, AD_p + 3$ |
| 5         | $AD_p - 4, \dots, AD_p + 4$ |

Table 5.1: Expansions of the Adaptive Cohort Fix-and-Optimize Matheuristic.

per iteration has to adapt to the expansion level to account for required additional solving time while simultaneously limiting wasting time for hard-to-solve problems. Taking into consideration the available laptop as well as the different benchmark instances, we define the time limit per iteration according to Equation 5.1.

$$TL_i = \min(RT, EL * T_v + T_f) \quad (5.1)$$

$TL_i$  represents the time limit per iteration, which is calculated as the minimum value between the remaining time of the experiment  $RT$  and the time limit resulting from the expansion level  $EL$  times a variable time part  $T_v$  plus a fixed time part  $T_f$ .

### 5.3 Application to Illustrative Example

To visualize how the Adaptive Cohort Fix-and-Optimize Matheuristic works, we present an illustrative example. The example is kept very small to focus on the important aspects of the matheuristic.

| Rooms | Number of beds | Specialism      |
|-------|----------------|-----------------|
| $r_1$ | 2              | General Surgery |
| $r_2$ | 2              | Cardiology      |
| $r_3$ | 1              | General Surgery |
| $r_4$ | 1              | Cardiology      |

Table 5.2: Rooms of the illustrative example.

The example contains eight patients, four rooms with one or two beds, and two specialisms. The individual properties of the rooms and patients are shown in Table 5.2 and Table 5.3. The planning horizon is four days long and the example comprises penalties for transfer (11), gender policy violation (5), room type preference (0.8) and specialism mismatch (1). The penalty weights in brackets are chosen identically to the values published on the PAS website (Demeester, Peter 2009).

| Patient | Admission | Length of stay | Gender | Pref. room type | Specialism      |
|---------|-----------|----------------|--------|-----------------|-----------------|
| $p_A$   | 1         | 3              | m      | 1               | Cardiology      |
| $p_B$   | 1         | 1              | m      | 2               | Cardiology      |
| $p_C$   | 2         | 2              | f      | 1               | Cardiology      |
| $p_D$   | 2         | 3              | m      | 1               | General Surgery |
| $p_E$   | 3         | 2              | f      | 2               | General Surgery |
| $p_F$   | 3         | 1              | f      | 1               | General Surgery |
| $p_G$   | 4         | 1              | m      | 1               | General Surgery |
| $p_H$   | 4         | 1              | f      | 1               | General Surgery |

Table 5.3: Patients of the illustrative example.

The first step is to create a feasible initial solution. The greedy assignment used to create the initial solution inserts the patients from longest to shortest length of stay (LoS) into the first available room. Additionally, the gender pairings are also taken into consideration. In this example, patient  $p_A$  and  $p_D$  are assigned first because of their LoS of 3. Patient  $p_A$  gets assigned to the first available room  $r_1$ . Since room  $r_1$  contains two beds, and patient  $p_A$  and  $p_D$  have the same gender, patient  $p_D$  gets assigned to room  $r_1$  as well. Next, patient  $p_C$  and  $p_E$  are assigned. As room  $r_1$  is already at full capacity, patient  $p_C$  is assigned to room  $r_2$ . Patient  $p_E$  is assigned to room  $r_2$  as well because the room still has free capacity and the genders match. The final patients to assign are  $p_B$ ,  $p_F$ ,  $p_G$  and  $p_H$ . Patient  $p_B$  is assigned to room  $r_1$  as the gender is the same as  $p_A$ 's and there is free capacity for its LoS as patient  $p_D$  only arrives on day 2. Patient  $p_F$  is assigned to room  $r_3$  because  $r_1$  and  $r_2$  are at full capacity during  $p_F$ 's stay. Patient  $p_G$  is joining  $p_D$  in room  $r_1$  and patient  $p_H$  is assigned to room  $r_2$  with  $p_E$  due to the gender policy.

| Rooms | Day 1      | Day 2      | Day 3      | Day 4      |
|-------|------------|------------|------------|------------|
| $r_1$ | $p_A, p_B$ | $p_A, p_D$ | $p_A, p_D$ | $p_D, p_G$ |
| $r_2$ |            | $p_C$      | $p_C, p_E$ | $p_E, p_H$ |
| $r_3$ |            |            | $p_F$      |            |
| $r_4$ |            |            |            |            |

Table 5.4: Initial solution, assignment of the illustrative example.

| Patients | Penalties |
|----------|-----------|
| $p_A$    | 5.4       |
| $p_B$    | 1         |
| $p_C$    | 1.6       |
| $p_D$    | 2.4       |
| $p_E$    | 2         |
| $p_F$    | 0         |
| $p_G$    | 0.8       |
| $p_H$    | 1.8       |
| Total    | 15.0      |

Table 5.5: Initial solution, patient penalties of the illustrative example.

When we look at the complete initial solution in Table 5.4, we can see that the rooms with low room numbers are at high capacity, as other rooms are kept completely empty. This is intended to provide room for big improvements early on. Additionally, the initial solution does not contain any transfers or gender policy violations, which have high penalty

weights. With this setup, we get a total penalty value of 15 as illustrated in Table 5.5. We take 5% (rounded up) of the planning horizon to build the subset of worst admission dates, which equals one admission date per iteration in this example. Patient  $p_A$  has the highest penalty value with 5.4 in the initial solution, as he is assigned to a bigger room than he would prefer with the wrong specialism for three days. Therefore, we group all patients with the same admission date as patient  $p_A$  to a cohort and keep this group of patients open for optimization. All other patients are fixed in their current rooms. Patient  $p_A$  shares the admission date with patient  $p_B$ , so we optimize  $p_A$  and  $p_B$  while keeping the rest of the assignment fixed. The optimal room for patient  $p_A$  is  $r_4$ , where his penalty value becomes zero because he gets a single room with the correct specialism. Patient  $p_B$  is now assigned to room  $r_2$ , which satisfies his specialism requirement. After the first iteration, we get the optimized patient assignment presented in Table 5.6 and a total penalty value of 8.6 as calculated in Table 5.7.

| Rooms | Day 1 | Day 2 | Day 3      | Day 4      |
|-------|-------|-------|------------|------------|
| $r_1$ |       | $p_D$ | $p_D$      | $p_D, p_G$ |
| $r_2$ | $p_B$ | $p_C$ | $p_C, p_E$ | $p_E, p_H$ |
| $r_3$ |       |       | $p_F$      |            |
| $r_4$ | $p_A$ | $p_A$ | $p_A$      |            |

Table 5.6: Assignment after the first and second iteration with no expansion.

| Patients | Penalties |
|----------|-----------|
| $p_A$    | 0         |
| $p_B$    | 0         |
| $p_C$    | 1.6       |
| $p_D$    | 2.4       |
| $p_E$    | 2         |
| $p_F$    | 0         |
| $p_G$    | 0.8       |
| $p_H$    | 1.8       |
| Total    | 8.6       |

Table 5.7: Patient penalties after the first and second iteration with no expansion.

The admission window expands once 50% of the admission dates have been investigated. This means that there is one more optimization to be done in this example before the first expansion. Hence, we look at the second highest penalty value of the initial solution in Table 5.5, which is the value 2.4 of patient  $p_D$ . We group the patients with admission date 2, which are patient  $p_C$  and  $p_D$ , and keep every other assignment fixed. In this case, there is no more improvement possible since patient  $p_D$  can't move to another room because of capacity issues, and patient  $p_C$  could only move to room  $r_1$  with the wrong specialism and a possible gender policy violation because of patient  $p_D$ .

After all iterations without expansion took place, the penalties of the current solution are calculated to determine the new admission date cohorts to investigate. Now, the admission window is expanded to prevent the matheuristic from repeating non-improving optimizations, which could ultimately lead to the matheuristic getting stuck in a local minimum. If we optimize the single highest penalty admission date again without any expansion, we would optimize patient  $p_C$  and  $p_D$  once again with no improvement. Therefore, in the first expansion, we add the following day to the admission window. From the

penalties of the current solution, patient  $p_D$  has the highest penalty value with 2.4. This means that the admission dates 2 and 3 are chosen to form the first cohort of patients to be optimized in this iteration. The expansion enables patient  $p_D$  to change rooms because patient  $p_F$  is also free for optimization. This swap increases the penalty of patient  $p_F$  by 0.8 but it eliminates the whole penalty for  $p_D$ , thus it results in a better solution overall as seen in Table 5.9.

| Rooms | Day 1 | Day 2 | Day 3      | Day 4      |
|-------|-------|-------|------------|------------|
| $r_1$ |       |       | $p_F$      | $p_G$      |
| $r_2$ | $p_B$ | $p_C$ | $p_C, p_E$ | $p_E, p_H$ |
| $r_3$ |       | $p_D$ | $p_D$      | $p_D$      |
| $r_4$ | $p_A$ | $p_A$ | $p_A$      |            |

Table 5.8: Assignment after the first iteration with one expansion.

| Patients | Penalties |
|----------|-----------|
| $p_A$    | 0         |
| $p_B$    | 0         |
| $p_C$    | 1.6       |
| $p_D$    | 0         |
| $p_E$    | 2         |
| $p_F$    | 0.8       |
| $p_G$    | 0.8       |
| $p_H$    | 1.8       |
| Total    | 7         |

Table 5.9: Patient penalties after the first iteration with one expansion.

For the second iteration after the first expansion, the admission dates 3 and 4 are selected because patient  $p_E$  has the second-highest penalty (Table 5.7). In this iteration,  $p_E$  and  $p_H$  move to room  $r_1$  to receive their preferred specialism, while patient  $p_G$  moves to room  $r_4$  to satisfy his single room preference although losing his specialism preference. The total penalty value after the set of iterations in the first expansion decreases to 4.2 (Table 5.11).

| Rooms | Day 1 | Day 2 | Day 3      | Day 4      |
|-------|-------|-------|------------|------------|
| $r_1$ |       |       | $p_E, p_F$ | $p_E, p_H$ |
| $r_2$ | $p_B$ | $p_C$ | $p_C$      |            |
| $r_3$ |       | $p_D$ | $p_D$      | $p_D$      |
| $r_4$ | $p_A$ | $p_A$ | $p_A$      | $p_G$      |

Table 5.10: Assignment after the second iteration with one expansion.

| Patients | Penalties |
|----------|-----------|
| $p_A$    | 0         |
| $p_B$    | 0         |
| $p_C$    | 1.6       |
| $p_D$    | 0         |
| $p_E$    | 0         |
| $p_F$    | 0.8       |
| $p_G$    | 1         |
| $p_H$    | 0.8       |
| Total    | 4.2       |

Table 5.11: Patient penalties after the second iteration with one expansion.

The second expansion adds the day before the admission date to the current admission window. Since the fundamental mechanism of the matheuristic is explained within the previous iterations, we abstain from continuing this illustrative example.

## 6. A Multi-Strategy Fix-and-Optimize Matheuristic (Oehrli)

The exact base model presented in Chapter 4 cannot solve large benchmark instances within a reasonable time. In this chapter, we present a matheuristic that addresses this limitation by decomposing the problem into small subproblems, each of which is solved exactly. This approach produces high-quality solutions within a fixed time budget. Section 6.1 provides an overview of the procedure. Section 6.2 describes the construction of the subproblems, the derivation of the parameters, and the group-construction strategies. Finally, Section 6.3 demonstrates the matheuristic using an illustrative example.

### 6.1 Overview of Matheuristic

The matheuristic follows the Fix-and-Optimize paradigm of Turhan and Bilgen (2017). In each iteration, a subset of patients is left free for reoptimisation while all other assignment decisions are fixed. The resulting subproblem has the same structure as the mixed-integer linear program of Chapter 4, restricted to the free patients. The procedure consists of three consecutive phases: a construction phase that builds a feasible initial solution, an improvement phase that reduces its cost through three alternating decomposition strategies, and a repair phase that targets the residual gender violations.

The construction phase builds the initial solution with a greedy heuristic that assigns each patient a single room for the entire stay, so that no transfer penalty is incurred at the outset. After the improvement phase, the repair phase reoptimises the patients involved in the remaining gender violations together with candidate swap partners. The remainder of this section motivates the design of the three improvement strategies. Their precise construction is deferred to Section 6.2.

The three decomposition strategies target complementary sources of inefficiency. The *room cluster* strategy groups all patients currently assigned to the same department, since these patients compete for the same rooms, and any mutual improvement requires solving them jointly. It is most effective when the construction phase has left a high departmental cost. The *time window* strategy optimizes all patients whose stays overlap with a short sequence of consecutive days. It is designed to address cross-period conflicts that cannot be captured by departmental grouping alone, since such conflicts may involve patients from different departments. The *swap chain* strategy constructs a chain of patients who



mutually block each other from moving to cheaper rooms and reoptimizes the chain jointly. It uncovers global blockings of the form “ $p_1$  would prefer the room of  $p_2$ , who would prefer the room of  $p_3$ ” that neither departmental nor temporal decomposition can resolve.

The construction phase eliminates one of the costliest sources of penalty by design: since each patient is assigned a single room for the entire stay, no transfer penalty is incurred in the initial solution.

The procedure is summarised in Algorithm 2. The time budget is split: a fraction of the total time  $T_{\text{total}}$  is reserved for the improvement phase and the remainder for the repair phase. The parameters that govern the subproblem size are derived automatically from the instance and are specified in Section 6.2. The concrete values used for the time-budget split and the subproblem-size parameters are reported in Section 7.2.

---

**Algorithm 2** Multi-Strategy Fix-and-Optimize Matheuristic

---

```

1: bestSolution = BuildGreedy()
2: bestCost = ObjValue(bestSolution)
3: Initialize strategy list, stall counter, and shuffle flag
4: while time remaining and stall counter below limit do
5:   Select decomposition strategy (advances only after a stalled pass)
6:   Build groups from bestSolution according to strategy
7:   for all groups in current decomposition do
8:     Solve the sub-problem
9:     if newCost < bestCost then
10:       bestSolution = newSolution
11:       bestCost = newCost
12:   if solution improved in this pass then
13:     Reset stall counter and shuffle flag
14:   else
15:     Increase stall counter and advance to next strategy
16:     if all strategies have stalled then
17:       Enable shuffle for next round
18: Apply gender repair to bestSolution
19: return bestSolution

```

---

## 6.2 Subproblems

The initial solution is constructed by a greedy heuristic. Patients are sorted in decreasing order of a priority score that combines the length of stay with the minimum attainable assignment cost  $\min_{r \in R} C_{p,r}$ . This prioritization gives precedence to patients who occupy capacity for many days and are costly to place in unsuitable rooms, so that the hardest placements are committed first while the room inventory is still wide. Each patient is then assigned to the cheapest room that respects both the capacity and the gender restrictions

on every day of the stay. If no such room exists, the heuristic falls back to the globally cheapest feasible room, which may incur a room-gender penalty. Such residual gender violations are treated as soft constraints in the objective and are subsequently targeted by the repair phase. Capacity violations do not occur on any benchmark instance considered in this thesis.

Because every patient receives a single room for the entire stay, no transfers occur in the initial solution. This eliminates the highest-weighted soft constraint by design and provides the improvement phase with a clean starting point.

Four parameters govern the subproblem size. They are derived automatically from the instance so that the matheuristic scales without manual tuning. They reflect the trade-off between the size of a subproblem and the number of subproblems that can be solved within the time budget. A larger subproblem captures more interactions between patients but takes longer to solve, so that fewer iterations are possible. The chosen values keep each subproblem small enough to be solved to a near-zero gap within seconds. The *candidate room set* restricts, for each patient, the rooms a subproblem may consider to those with the lowest static assignment cost. This is a heuristic reduction rather than a dominance rule: an excluded room could in principle still be part of a globally optimal solution, so the restriction may forgo optimality in exchange for smaller subproblems. The *group size* determines how many patients are reoptimised jointly and is reduced on instances where the average daily patient density is very high. The *subproblem time limit* increases with instance size. Each subproblem is supplied with the current best solution as a warm start. The concrete values of these parameters are reported in Section 7.2.

In the improvement phase, three strategies are used. The *room cluster* strategy groups patients according to the department of their currently assigned room. These patients compete for the same room capacities and therefore form natural candidates for room exchanges. Within each department, patients are sorted by their individual costs contribution and partitioned into chunks of the predefined group size. The *time window* strategy slides a window of consecutive days across the planning horizon. The window length is set to approximately one-third of the horizon, capped at fourteen days, and successive windows overlap by 50 % so that every day falls inside at least two windows. For each window, the patients with the highest assignment cost within that period are selected as the free set. Only their assignments are reoptimized.

The *swap chain* strategy identifies groups of patients who mutually block each other from moving to cheaper rooms. Starting from the most expensive patient, a breadth-first search adds any patient who currently occupies a room that would be cheaper for the most expensive patient and whose stay overlaps with it. The search continues until the group reaches the predefined size. The process is then repeated for the next most expensive patient not yet contained in a chain.

The three strategies alternate in a fixed order. Whenever a strategy completes a full pass over its groups without improving the incumbent, the pass is counted as a stall, and the next strategy is activated. Any improvement resets the stall counter. After every complete pass over all three strategies without improvement, the swap chain order is randomized to diversify the search. The improvement phase terminates once three

consecutive complete passes over all strategies have stalled, or once the improvement time budget  $T_1$  is exhausted.

The repair phase targets the remaining mixed-gender occupancies in no-mixing rooms. In each iteration, the patients currently involved in those room-days are collected into a set of affected patients, which is then augmented with candidate swap partners in three stages of increasing relaxation: first candidates of the same gender from the same department, then candidates of the same gender from any department, and finally, any patient, regardless of gender or department. One subproblem is solved per stage over the resulting union. The solution is accepted only if it strictly reduces the number of mixed room-days and increases the objective by less than two percent of the current best value. As soon as a stage succeeds, the loop continues to the next iteration. If all three stages fail, the repair loop terminates, and any remaining violations are reported.

### 6.3 Application to Illustrative Example

We apply the matheuristic to a small handcrafted instance to make the steps of Algorithm 2 concrete. This instance is not representative of the large benchmark instances for which the matheuristic was designed. Performance on those is reported in Chapter 7. The instance has a planning horizon of seven nights, five rooms, and eighteen elective patients. It comprises three departments, each with one major specialism: cardiology ( $d_1$ ), pediatrics ( $d_2$ ), and general medicine ( $d_3$ ). The rooms and the patients are described in Tables 6.1 and 6.2.

| Room  | Dept. | Cap. | Gender | Age limit | Features          |
|-------|-------|------|--------|-----------|-------------------|
| $r_1$ | $d_1$ | 2    | mixed  | none      | oxygen            |
| $r_2$ | $d_1$ | 1    | mixed  | none      | oxygen, telemetry |
| $r_3$ | $d_2$ | 2    | mixed  | $\leq 16$ | oxygen            |
| $r_4$ | $d_3$ | 2    | mixed  | none      | oxygen            |
| $r_5$ | $d_3$ | 3    | mixed  | none      | none              |

Table 6.1: Rooms of the illustrative instance.

The greedy heuristic produces a starting solution with objective 148.4. Of this, 35.0 are gender penalties from seven mixed-gender room-days in  $r_1$  and  $r_3$  (weight 5.0 each), and the remaining 113.4 are assignment penalties ( $C_{p,r}$ ). No transfers occur. The cost concentrates in a few patients who cannot be placed in a room that matches both their specialism and their medical requirements: several cardiology and pediatric patients are displaced into the general medicine ward  $r_5$ , which offers neither the required oxygen equipment, nor the matching specialism, and one general medicine patient is placed in a cardiology room with the wrong specialism.

Table 6.3 traces the run from the greedy start to the final solution. The *room cluster* strategy forms a single group of the general medicine patients, which is the only department meeting the minimum size of eight, and removes the gender mixing and specialism

| Patient  | Gender | Age | Specialism    | Room pref. | Req. equip. | Adm. | LoS |
|----------|--------|-----|---------------|------------|-------------|------|-----|
| $p_1$    | F      | 55  | Cardiology    | twin       | oxygen      | 0    | 3   |
| $p_2$    | M      | 62  | Cardiology    | single     | telemetry   | 0    | 4   |
| $p_3$    | M      | 70  | Cardiology    | twin       | oxygen      | 2    | 3   |
| $p_4$    | F      | 48  | Cardiology    | twin       | oxygen      | 3    | 3   |
| $p_5$    | M      | 58  | Cardiology    | twin       | oxygen      | 4    | 3   |
| $p_6$    | F      | 8   | Pediatrics    | twin       | oxygen      | 0    | 4   |
| $p_7$    | M      | 12  | Pediatrics    | twin       | oxygen      | 1    | 3   |
| $p_8$    | F      | 14  | Pediatrics    | twin       | oxygen      | 4    | 2   |
| $p_9$    | M      | 10  | Pediatrics    | twin       | oxygen      | 5    | 2   |
| $p_{10}$ | F      | 35  | Gen. Medicine | ward       | none        | 0    | 3   |
| $p_{11}$ | F      | 42  | Gen. Medicine | ward       | none        | 1    | 3   |
| $p_{12}$ | F      | 50  | Gen. Medicine | twin       | none        | 2    | 3   |
| $p_{13}$ | M      | 60  | Gen. Medicine | ward       | none        | 0    | 3   |
| $p_{14}$ | M      | 38  | Gen. Medicine | ward       | none        | 3    | 3   |
| $p_{15}$ | M      | 45  | Gen. Medicine | ward       | none        | 5    | 2   |
| $p_{16}$ | F      | 28  | Gen. Medicine | twin       | none        | 4    | 3   |
| $p_{17}$ | M      | 52  | Gen. Medicine | ward       | none        | 1    | 4   |
| $p_{18}$ | F      | 33  | Gen. Medicine | ward       | none        | 4    | 2   |

Table 6.2: Patients of the illustrative instance. *Adm.* is the admission night. *LoS* is the length of stay in nights.

mismatches within that department, yielding a drop of 51.0. The cardiology and pediatric patients are not touched in this pass because their departments are too small to form a cluster. The *time window* strategy reaches them across two overlapping windows,  $[2, 6)$  and  $[4, 7)$ , which together cover the peak occupancy period and reassign these patients to appropriate rooms, contributing a further 45.0. A subsequent *room cluster* pass finds an additional 12.6 once the temporal rearrangements have changed the departmental cost landscape, because the time window pass has moved patients between departments, so that the departmental groups now differ from those of the first *room cluster* pass and expose further exchanges. The *swap chain* strategy stalls on this instance, since it can only resolve cases in which patients mutually block each other from moving to cheaper rooms, whereas the residual cost here is structural gender mixing in the cardiology and pediatric rooms rather than the result of inter-patient blockings.

The repair phase then targets the seven remaining mixed-gender room-days, which carry a gender penalty of 35.0. All three expansion stages fail: the tight bed utilization of the instance leaves no reassignment that reduces the number of mixed room-days without exceeding the two-per-cent regression threshold. The matheuristic therefore terminates with the objective reached at the end of the improvement phase.

The example illustrates the complementary roles of the three strategies: *room cluster* corrects departmental inefficiencies, *time window* eliminates cross-period conflicts that departmental grouping cannot reach, and the *swap chain* targets the inter-patient block-

| Iteration                              | Strategy     | Groups | Improvement | Incumbent   |
|----------------------------------------|--------------|--------|-------------|-------------|
| —                                      | greedy start | —      | —           | 148.4       |
| 1                                      | room cluster | 1      | 51.0        | 97.4        |
| 3                                      | time window  | 2      | 45.0        | 52.4        |
| 6                                      | room cluster | 1      | 12.6        | 39.8        |
| Final objective (after repair phase)   |              |        |             | <b>39.8</b> |
| Relative improvement over greedy start |              |        |             | 73.2 %      |

Table 6.3: Trajectory of the matheuristic on the illustrative instance. Only the improving iterations are listed. The *Iteration* column gives the corresponding outer-loop iteration of Algorithm 2, and *Groups* the number of subproblems within that iteration that improved the incumbent. The repair phase leaves the objective unchanged, so the value after the improvement phase equals the final objective.

ings that arise in larger instances. The final objective of 39.8 consists of the residual gender penalties (35.0) and room size preference violations (4.8). All other penalty components are zero.

## 7. Experimental Analysis

This chapter presents the setup for the experimental analysis with the different parameter values as well as the results of the two matheuristics. In Section 7.1, we present the benchmark data sets used in the experiment. In Section 7.2, we introduce the experimental setup with its different parameter values. In Section 7.3, we discuss the results of the Adaptive Cohort Fix-and-Optimize Matheuristic. In Section 7.4, we discuss the results of A Multi-Strategy Fix-and-Optimize Matheuristic.

### 7.1 Benchmark Data Sets

For the experimental analysis, the first 12 benchmark datasets from the PAS website (Demeester, Peter 2009) are used. We leave out dataset number 13 because this dataset is the only one that contains multiple specialisms for a patient during a stay. In Table 7.1 the properties of the datasets are presented. We can see that the size of the problems varies with the datasets, especially in terms of the planning horizon and the number of patients. As there is a department for every specialism, we abstain from tabulating the number of departments as it is equal to the number of specialisms.

| Dataset | Rooms | Beds | Room prop. | Specialisms | Patients | Planning horizon |
|---------|-------|------|------------|-------------|----------|------------------|
| 1       | 98    | 286  | 2          | 4           | 693      | 14               |
| 2       | 151   | 465  | 2          | 6           | 778      | 14               |
| 3       | 131   | 395  | 2          | 5           | 757      | 14               |
| 4       | 155   | 471  | 2          | 6           | 782      | 14               |
| 5       | 102   | 325  | 2          | 4           | 631      | 14               |
| 6       | 104   | 313  | 2          | 4           | 726      | 14               |
| 7       | 162   | 472  | 4          | 6           | 770      | 14               |
| 8       | 148   | 441  | 4          | 6           | 895      | 21               |
| 9       | 105   | 310  | 4          | 4           | 1400     | 28               |
| 10      | 104   | 308  | 4          | 4           | 1575     | 56               |
| 11      | 107   | 318  | 4          | 4           | 2514     | 91               |
| 12      | 105   | 310  | 4          | 4           | 2750     | 84               |

Table 7.1: Datasets of the experimental analysis.

## 7.2 Experimental Setup

To test the performance of our matheuristics, we compare the objective function value of the matheuristic with the objective function value of the base model after the same amount of time. The time limit for this thesis is set to one hour. To ensure the comparability of the results for the basic model and the matheuristic, the test runs for each matheuristic are executed on the same computer under the same circumstances. As we do not compare the matheuristics with one another, there is no need to use the same computer for both matheuristics. The Adaptive Cohort Fix-and-Optimize Matheuristic is run on a MacBook Air with a Dual-Core Intel Core i5 1.6 GHz processor and 8GB of RAM. For the Multi-Strategy Fix-and-Optimize Matheuristic, a Lenovo Thinkpad X1 with an Intel Core i7 2.1 GHz processor and 32 GB of RAM is used. The model and the matheuristics are programmed using Python, and the chosen solver is gurobi 13.0.1.

To calculate the penalty values, we use the same penalty weights as proposed on the PAS website (Demeester, Peter 2009), which are illustrated in Table 7.2.

| Soft Constraint              | Penalty |
|------------------------------|---------|
| Specialism of the department | 1.0     |
| Specialism of the room       | 1.0     |
| Gender policy                | 5.0     |
| Age policy                   | 10.0    |
| Mandatory room features      | 5.0     |
| Preferred room features      | 2.0     |
| Room type                    | 0.8     |
| Transfers                    | 11.0    |

Table 7.2: Penalty weights of the experimental analysis.

For the penalty calculation, we have to take a few assumptions that might be different from Turhan and Bilgen (2017). First, we only penalize gender-mixed rooms that are not supposed to be gender-mixed. This means we do not penalize an F room if there are only male patients in there, as the binary variable in the model only accounts for mixed rooms. By using this variable, a distinction between a purely female or purely male room would not be possible, and from our point of view, it does not make sense why rooms should only be allowed for one specific gender. As stated on the PAS website (Demeester, Peter 2009), we multiply the penalty for the room size by the difference between the assigned and the preferred room size. As a base for this calculation, we choose the exact values in the data where they use the values 1, 2, and 4 for the different room types. Since the department specialisms as well as the room specialisms are split into first and second choice specialisms, we follow the same approach as with the room size and penalize first choice, second choice, and no specialism based on their difference. Departments or rooms with a first choice specialism receive a penalty with the factor 0, second choice specialisms are penalized with the factor 1, and if the department or room does not have the required

specialism at all, the penalty is incurred with a factor 2.

| Parameter | Parameter value |
|-----------|-----------------|
| $CS$      | 5%              |
| $PL$      | 50%             |
| $T_v$     | 120s            |
| $T_f$     | 60s             |

Table 7.3: Parameter values for the experimental analysis of the Adaptive Cohort Fix-and-Optimize Matheuristic.

In Table 7.3, we present the parameter values used in the experimental analysis of the Adaptive Cohort Fix-and-Optimize Matheuristic. We set the cohort size  $CS$  to 5% of the planning horizon and the planning limit  $PL$  to 50% so that half of the admission dates are explored per set of iterations. To determine the time limit per iteration, we define variable and fixed time parts  $T_v$  and  $T_f$  with values of 120s and 60s.

For the Multi-Strategy Fix-and-Optimize Matheuristic, the four parameters that govern the subproblem size are not set to fixed values but are derived automatically from each instance, so that the matheuristic scales without manual tuning. This automatic derivation applies only to this matheuristic. The values are fixed after preliminary experiments on a subset of the benchmark instances and are not subjected to a systematic parameter study. They reflect the trade-off between the size of a subproblem and the number of subproblems that can be solved within the time budget. Of the total time budget, a fraction  $\alpha = 0.85$  is reserved for the improvement phase and the remaining  $1 - \alpha$  for the repair phase. The candidate room set is restricted to roughly 25 % of all rooms with the lowest static assignment cost per patient. The group size is set to approximately 15 % of the total bed count, reduced on instances where the average daily patient density is very high, and clamped between 25 and 60 patients. The subproblem time limit increases with instance size, ranging from 20 s for small instances to 45 s for large ones. Each subproblem is supplied with the current best solution as a warm start and solved with a MIP gap tolerance of 0.5 %.

### 7.3 Discussion of the Adaptive Cohort Fix-and-Optimize Matheuristic (Szabo)

In Table 7.4, the results of the experimental analysis of the Adaptive Cohort Fix-and-Optimize Matheuristic, applied to the 12 benchmark instances, are presented.



| Instance   | Base obj. | Gap <sub>base</sub> | Heur.obj. | Heur. vs. Base |
|------------|-----------|---------------------|-----------|----------------|
| testdata1  | 2,398.6   | 1.96%               | 2,575.6   | +7.38%         |
| testdata2  | 4,995.6   | 11.59%              | 4,811.2   | −3.69%         |
| testdata3  | 2,755.8   | 2.55%               | 2,974.8   | +7.95%         |
| testdata4  | 4,776.4   | 12.96%              | 4,708.0   | −1.43%         |
| testdata5  | 2,051.0   | 1.90%               | 2,086.0   | +1.71%         |
| testdata6  | 3,096.0   | 5.69%               | 3,132.2   | +1.17%         |
| testdata7  | 3,322.6   | 0.39%               | 3,379.8   | +1.72%         |
| testdata8  | 10,335.6  | 21.21%              | 9,578.0   | −7.33%         |
| testdata9  | 142,906.0 | -                   | 37,485.8  | −73.77%        |
| testdata10 | 144,015.0 | -                   | 17,444.6  | −87.89%        |
| testdata11 | 248,756.0 | -                   | 29,136.2  | −88.29%        |
| testdata12 | 276,337.0 | -                   | 46,416.4  | −83.20%        |

Table 7.4: Results of the Adaptive Cohort Fix-and-Optimize Matheuristic. Dashes represent missing MIP Gaps of the base model, because no objective bound was found.

For the small instances 1-7 with a planning horizon of 14 days, we observe mixed results. While the matheuristic produces a solution with an objective function value approximately 8% higher than the base model for instances 1 and 3, the final solution for instances 5, 6, and 7 gets within 2% of the base model’s result. For instances 2 and 4, the matheuristic outperforms the base model by 3.69% and 1.43%, respectively. For the latter two instances, we can observe a higher MIP gap of approximately 10% compared to the other small instances. This indicates that the base model optimization would need more runtime to close the optimality gap.

In instance 8, the planning horizon is increased to 21 days, challenging the scalability of the matheuristic. In this medium-sized instance, the matheuristic outperforms the base model by 7.33%. However, the base model was still able to start the optimization process and produce a reasonable solution within the one-hour time limit. For the large instances 9-12, the base model needs too much time to presolve and is therefore not able to produce better feasible solutions within the time limit. Hence, it ends up with the only feasible solution found, which is the initial solution with its large objective function value. In contrast, the matheuristic manages to produce an improved solution, undercutting the initial solution of the base model by 73.77% to 88.29%. This large difference highlights the scalability of the matheuristic.

To further investigate the performance of the matheuristic, we perform a convergence analysis, comparing the current best solution over time to the best final solution across both methods. For comparison across different instances, we calculate the relative gap according to Equation 7.1. The relative gap shows the progress of the optimization by indicating the remaining gap to the reference solution in percent after each iteration. We

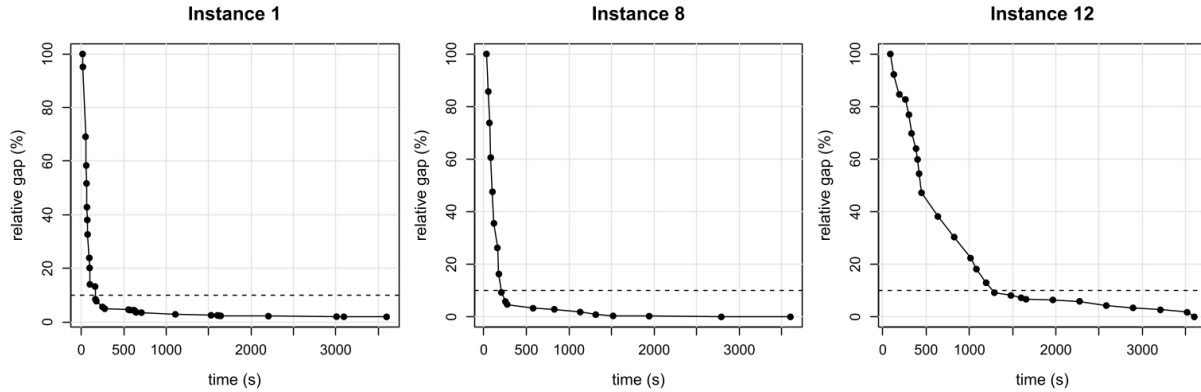


Figure 7.1: Convergence curve of the matheuristic for three different-sized instances (instance 1: small, instance 8: medium, instance 12: large) over the one-hour time limit. The relative gap represents the remaining gap to the best solution across both approaches relative to the initial gap. The points represent the result at the end of each iteration. The dotted line at 10% relative gap shows when the matheuristic reached 90% of the total improvement.

use the best final solution across both, the matheuristic  $S_m$  and the base model  $S_b$  as a reference solution for comparison. For normalization, we calculate the initial gap by subtracting the reference solution from the initial solution  $S_i$ . The current gap is the result of the difference between the current solution  $S_c$  and the reference solution. The relative gap is then obtained by dividing the current gap by the initial gap, representing the optimization progress during the experiment.

$$Gap_r = \frac{S_c - \min(S_m, S_b)}{S_i - \min(S_m, S_b)} \quad (7.1)$$

We select three representative instances of different sizes (small, medium, and large planning horizons) to perform this analysis while covering cases where the matheuristic performs better and worse than the exact base model. The convergence curves in Figure 7.1 show typical matheuristic behavior with fast early improvements, gradual refinement, and final stabilization. Comparing the starting points of the three instances, the largest instance requires slightly longer to obtain the initial solution. However, this difference is negligible for a time limit of one hour. For small and medium-sized instances, the matheuristic rapidly converges to a reasonably good solution, covering 90% of the total improvement within approximately 200s. The first few iterations contribute most to the improvement. After this initial great improvement phase, the matheuristic improves the solution only marginally in the remaining iterations. This shows that the chosen time limit could be reduced drastically for the smaller instances without restricting the performance of the matheuristic. In contrast, the rate of improvement is slower for the large instance, and the convergence curve decreases more gradually. This shows the complexity of the instance, requiring more iterations before reaching a plateau. While most improvements

still happen in the early optimizations, later iterations also contribute by a noticeable amount, leading to an overall flatter convergence curve. We also see a noticeable improvement in the last iteration, which indicates that the matheuristic has not yet reached its full improvement potential within the given time limit. Overall, the matheuristic consistently improves the objective function value for all analyzed instances and reaches good solutions during the selected time limit, showing good scalability.

## 7.4 Discussion of A Multi-Strategy Fix-and-Optimize Matheuristic (Oehrli)

In this section, we discuss the experimental results of the multi-strategy Fix-and-Optimize matheuristic developed in Chapter 6. The matheuristic is benchmarked against the base model of Turhan and Bilgen (2017) from Chapter 4 on the twelve instances introduced in Section 7.1. Both approaches operate under identical conditions, in particular under the same wall-clock budget of one hour per instance, as specified in Section 7.2.

The principal finding of the experiments is that the matheuristic reaches an acceptable solution very quickly. Across all twelve benchmark instances, the matheuristic reaches a solution within twenty percent of the final base-model objective in approximately one minute on average (and within four minutes for every instance), and within fifteen percent in approximately three minutes on average. On ten of the twelve instances, the best-known solution reaches a gap of ten percent within roughly six minutes, and on seven of the twelve instances, a gap of five percent within roughly eight minutes. The size of these gains depends on the congestion of the instance. On instances where the room inventory provides enough variety for most patients to be placed in their department’s primary room without conflict, the assignment costs are already low after the construction phase, and the improvement phase finds relatively little to correct. On large, heavily congested instances, the room scarcity forces the construction phase into many compromises, and the improvement phase provides substantially larger gains. On several large instances, in particular *testdata9*, *testdata11*, and *testdata12*, the matheuristic returns a lower final best-known solution than the base model within the one-hour budget. This comparison refers to the final best-known solution of the base model after one hour, not to a proven optimum. The supporting evidence is presented in Figure 7.2 and in Table 7.5.

The convergence trajectories in Figure 7.2 confirm and refine this picture. For each instance, the figure shows the matheuristic best-known solution as a function of wall-clock time, with the final base model objective indicated by a dashed reference line where it lies within the plotted range. Two structural features are common to all twelve instances. First, the steepest decrease in the best-known solution occurs within the first three to five minutes. The curves drop sharply from the greedy starting solution and then bend visibly. Second, the curves remain monotonically decreasing throughout the budget rather than reaching a complete plateau, that is, improvements continue to accrue, although at a progressively diminishing rate. This pattern is consistent with the intended behavior of the Fix-and-Optimize scheme: the initial iterations expose the substantial slack that is present

in the greedy starting solution, while later iterations operate on a near-locally-optimal best-known solution in which only small improvements remain. The second feature shows that, although the bulk of the gain is captured early, the remainder of the time budget is not spent on a stagnant search, but on small, steady refinements.

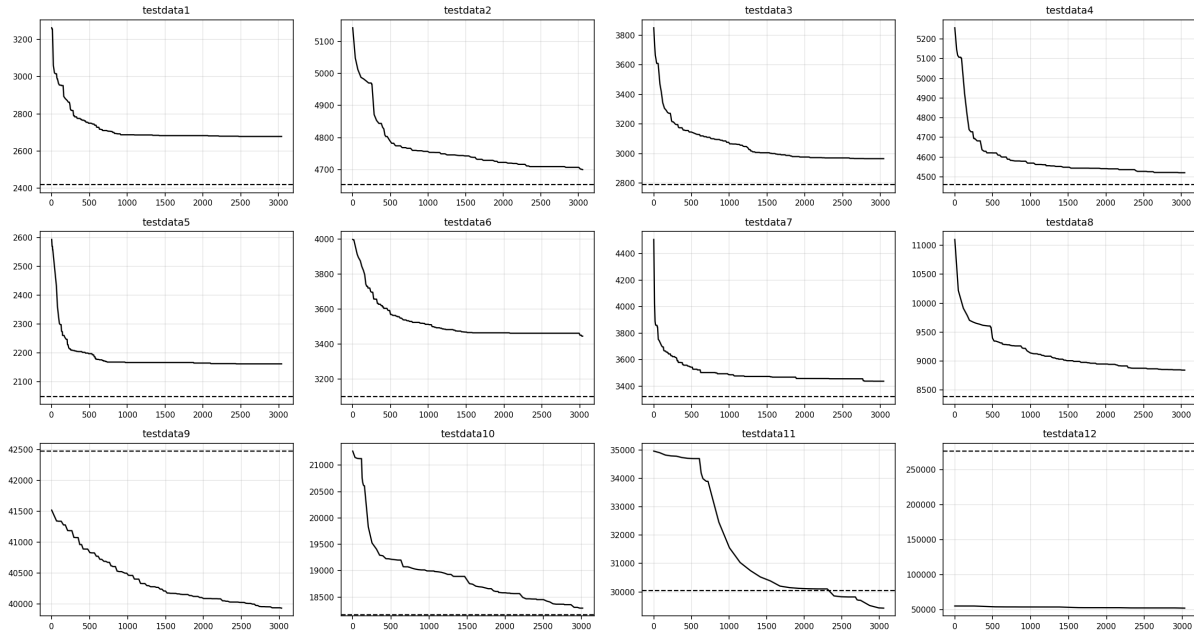


Figure 7.2: Convergence of the matheuristic over the one-hour time budget. The solid line shows the matheuristic best-known solution over time, and the dashed line indicates the final base model objective.

Table 7.5 reports the underlying numbers instance by instance. For each instance, the table gives the objective function value of the base model and of the matheuristic at the end of the one-hour budget, the optimality gap of the base model as returned by the solver, and the position of the matheuristic relative to the base model in percent, where a positive value means that the matheuristic is above the base model and a negative value means that it is below. The four right-most columns report the wall-clock time in seconds at which the matheuristic first reaches the twenty, fifteen, ten, and five percent quality thresholds. Entries marked “n.r.” indicate that the threshold was not reached within the budget. The mean row at the bottom is taken only over the instances on which the corresponding threshold was reached. The threshold times in Table 7.5 also show that the matheuristic reaches useful solution-quality levels early in the run. The twenty percent threshold is reached on all twelve instances, with a mean time of less than one minute among the reached instances. The fifteen percent threshold is also reached on all instances, after roughly three minutes on average. The ten percent threshold is reached on ten instances, and the five percent threshold on seven instances. Thus, the table confirms that the matheuristic produces most of its practically relevant solution

| Instance            | Base obj. | Gap <sub>base</sub> | Heur. obj. | Heur. vs. base | $t_{20\%}$ | $t_{15\%}$ | $t_{10\%}$ | $t_{5\%}$ |
|---------------------|-----------|---------------------|------------|----------------|------------|------------|------------|-----------|
| testdata1           | 2,418.4   | 2.82 %              | 2,678.2    | +10.74 %       | 160.0      | 337.4      | n. r.      | n. r.     |
| testdata2           | 4,652.0   | 5.06 %              | 4,699.6    | +1.02 %        | 0.2        | 0.2        | 34.4       | 284.7     |
| testdata3           | 2,789.4   | 3.66 %              | 2,966.0    | +6.33 %        | 119.4      | 287.5      | 1,006.2    | n. r.     |
| testdata4           | 4,459.6   | 6.73 %              | 4,518.0    | +1.31 %        | 0.2        | 37.6       | 162.3      | 360.0     |
| testdata5           | 2,049.0   | 1.77 %              | 2,162.0    | +5.51 %        | 63.1       | 90.0       | 181.3      | n. r.     |
| testdata6           | 3,100.6   | 5.82 %              | 3,442.6    | +11.03 %       | 236.4      | 526.8      | n. r.      | n. r.     |
| testdata7           | 3,323.2   | 0.42 %              | 3,438.2    | +3.46 %        | 17.5       | 60.3       | 187.3      | 986.1     |
| testdata8           | 8,387.0   | 2.55 %              | 8,843.0    | +5.44 %        | 113.4      | 327.4      | 892.9      | n. r.     |
| testdata9           | 42,471.8  | 36.80 %             | 39,930.8   | −5.98 %        | 0.2        | 0.2        | 0.2        | 0.2       |
| testdata10          | 18,167.4  | 17.63 %             | 18,257.0   | +0.49 %        | 0.2        | 123.6      | 204.3      | 693.5     |
| testdata11          | 30,037.4  | 24.89 %             | 29,349.0   | −2.29 %        | 0.2        | 630.4      | 866.2      | 1,148.4   |
| testdata12          | 276,337.0 | 87.30 %             | 52,084.4   | −81.15 %       | 0.2        | 0.2        | 0.2        | 0.2       |
| Mean (reached only) |           |                     |            |                | 59.3       | 201.8      | 353.5      | 496.2     |

Table 7.5: Final objective function values of the base model and of the matheuristic after one hour, together with the wall-clock times in seconds at which the matheuristic first reaches the indicated quality threshold relative to the final base model value. A positive value in the column *Heur. vs. base* indicates that the matheuristic is above the base model at the end of the budget, a negative value indicates that it is below.

quality within the first few minutes, while the remaining time is mainly used for smaller refinements. Two observations on the contents of Table 7.5 deserve emphasis. The first concerns testdata12, the largest instance in the benchmark. The base model returns an objective function value of 276,337 with a remaining optimality gap above eighty-seven percent, whereas the matheuristic returns 52,084, an improvement of more than eighty percent within the same one-hour budget. The observed gaps suggest that the base model has difficulty improving its best-known solution within the one-hour time budget on the largest instances, whereas the decomposition into small subproblems insulates the matheuristic from this combinatorial explosion. A similar but less pronounced behavior is observed on testdata9 and testdata11, the other two large instances, on which the base model also fails to close its gap to a meaningful level. The second observation concerns the small and medium instances, where the situation is reversed. The base model reaches an optimality gap below six percent on six of the nine smaller instances within the hour, and on those instances, the matheuristic remains between one and eleven percent above the base model at the end of the budget. The instance on which the matheuristic is furthest from the base model, testdata1, is precisely the one on which the base model terminates with the smallest optimality gap, so that the absolute room for the matheuristic to improve within the budget is small from the outset.

A complementary sensitivity analysis on a medium-sized instance (testdata5) confirms that all three strategies are necessary: removing any single one worsens the final objective,

while on the extreme instances the dominant contributor shifts, with room clustering being most important on the smallest instance and swap chains on the largest, so that no two-strategy subset performs well across the whole benchmark. Taken together, the figure and the table support the use of the multi-strategy Fix-and-Optimize matheuristic as a complementary approach to the base model rather than as a strict replacement. On small and medium instances on which the base model can prove a small optimality gap within the hour, the matheuristic delivers a competitive solution in a small fraction of the time, but does not improve on the base model at the end of the budget. On the large instances on which the base model saturates the time limit without closing its gap, the matheuristic is the more reliable choice in absolute terms. The speed at which an acceptable solution is reached, within one minute, for a gap of twenty percent and within a few minutes for a gap of fifteen percent on every tested instance, is the main argument in favor of the matheuristic for any planning workflow in which a one-hour MIP run is not realistic in practice.

## 8. Conclusion and Outlook

Patient admission scheduling is a critical process in the healthcare sector to allocate available resources efficiently and increase patient satisfaction. Manual planning in such complex problems seems to be impossible. Hence, computer-based solution approaches are necessary. This thesis addresses the PAS problem in a static setting by implementing two different fix-and-optimize matheuristics.

The Adaptive Cohort Fix-and-Optimize Matheuristic uses penalty values to identify patients with a bad current assignment and optimizes the patient cohort with its common admission date. This matheuristic finds better objective function values in some of the smaller instances and only exceeded the exact optimization by 8 percent at most. For small and medium-sized problems the matheuristic rapidly identifies reasonably good solutions within the first few minutes. Furthermore, the matheuristic proves to have good scalability for more complex problems, since it provides better solutions in the four biggest instances where the exact optimization fails to produce solutions, because of the large computational power required.

The Multi-Strategy Fix-and-Optimize Matheuristic decomposes the problem into three alternating strategies: room clustering, time-window decomposition, and swap chains of mutually blocking patients. The matheuristic achieves solutions within twenty percent of the exact base model objective on all twelve benchmark instances, on average within roughly one minute, and outperforms the base model on several large instances within the one-hour time limit. The matheuristic, therefore, complements the exact approach rather than replacing it: the exact model remains preferable when it can prove a tight optimality gap, while the matheuristic is the more reliable choice when the problem size makes a full MIP run impractical.

When it comes to healthcare costs, factors like the increasing life expectancy do not hint towards a decrease in healthcare costs in the near future. Therefore, research for optimization in resource allocation in this sector will remain valuable for the economy as well as society in general. From the technical side, current trends in artificial intelligence and quantum computing could pave new ways for approaching optimization problems. Additionally, the PAS problem itself could be expanded with more features, for example, by differentiating the penalties for certain patients since in a real-world setting, some patients require to be treated differently based on their insurance level, e.g., when it comes to room size or expertise. Therefore, the problem could be expanded to better account for the ambiguity and uncertainty of the real world, which would further increase the practical value.

# Bibliography

- [1] Bastos, L. S. L., Marchesi, J. F., Hamacher, S., Fleck, J. L. (2019), A mixed integer programming approach to the patient admission scheduling problem. *European Journal of Operational Research* 273(3), 831–840
- [2] Bilgin, B., Demeester, P., Misir, M., Vancroonenburg, W., Vanden Berghe, G. (2012), One hyper-heuristic approach to two timetabling problems in health care. *Journal of Heuristics* 18(3), 401–434
- [3] Bolaji, A. L., Bamigbola, A. F., Shola, P. B. (2018), Late acceptance hill climbing algorithm for solving patient admission scheduling problem. *Knowledge-based systems* 145, 197–206
- [4] Burke, E. K., Bykov, Y. et al. (2008), A late acceptance strategy in hill-climbing for exam timetabling problems, in PATAT 2008 Conference, Montreal, Canada, pp. 1–7
- [5] Ceschia, S., Schaerf, A. (2011), Local search and lower bounds for the patient admission scheduling problem. *Computers & Operations Research* 38(10), 1452–1463
- [6] Ceschia, S., Schaerf, A. (2012), Modeling and solving the dynamic patient admission scheduling problem under uncertainty. *Artificial intelligence in medicine* 56(3), 199–205
- [7] Demeester, P., Souffriau, W., De Causmaecker, P., Vanden Berghe, G. (2010), A hybrid tabu search algorithm for automatically assigning patients to beds. *Artificial Intelligence in Medicine* 48(1), 61–70
- [8] Demeester, Peter (2009), Patient admission scheduling website. <<https://people.cs.kuleuven.be/~tony.wauters/wim.vancroonenburg/pas/>>; viewed May 30,2026.
- [9] Dueck, G. (1993), New optimization heuristics: The great deluge algorithm and the record-to-record travel. *Journal of Computational physics* 104(1), 86–92
- [10] Federal Statistical Office (2026a), Health care costs and financing in 2024, key figures. <<https://www.bfs.admin.ch/asset/de/36515499>>; viewed May 30,2026.
- [11] Federal Statistical Office (2026b), Health care costs and financing in 2024, press release. <<https://www.bfs.admin.ch/asset/en/36515474>>; viewed May 30,2026.



- [12] Glover, F. (1986), Future paths for integer programming and links to artificial intelligence. *Computers & operations research* 13(5), 533–549
- [13] Kifah, S., Abdullah, S. (2015), An adaptive non-linear great deluge algorithm for the patient-admission problem. *Information Sciences* 295, 573–585
- [14] Kirkpatrick, S., Gelatt Jr, C. D., Vecchi, M. P. (1983), Optimization by simulated annealing. *science* 220(4598), 671–680
- [15] Liu, H., Wang, Y., Hao, J.-K. (2024), Solving the patient admission scheduling problem using constraint aggregation. *European Journal of Operational Research* 316(1), 85–99
- [16] Lusby, R. M., Schwierz, M., Range, T. M., Larsen, J. (2016), An adaptive large neighborhood search procedure applied to the dynamic patient admission scheduling problem. *Artificial Intelligence in Medicine* 74, 21–31
- [17] Range, T. M., Lusby, R. M., Larsen, J. (2014), A column generation approach for solving the patient admission scheduling problem. *European Journal of Operational Research* 235(1), 252–264
- [18] Talbi, E.-G. (2009), *Metaheuristics: From Design to Implementation*. John Wiley & Sons
- [19] Turhan, A. M., Bilgen, B. (2017), Mixed integer programming based heuristics for the patient admission scheduling problem. *Computers & Operations Research* 80, 38–49
- [20] UBS (2026), UBS Worry Barometer 2025. <<https://www.gfsbern.ch/wp-content/uploads/2025/12/ubs-worry-barometer-2025-1.pdf>>; viewed May 30, 2026.

Statement of authorship for the Master's thesis entitled Optimizing Patient Admission Using Mathematical Programming from the year 2026

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